Adaptive Feature Pyramid Networks for Fine-Grained Image Classification

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Abstract

We propose Adaptive Feature Pyramid Networks (AFPN), a novel architecture for fine-grained image classification that dynamically adjusts feature fusion weights across pyramid levels. Unlike conventional FPN approaches that apply uniform top-down connections, AFPN introduces a lightweight channel-spatial attention gate at each lateral connection, enabling the network to selectively emphasize discriminative features for subtle inter-class differences. We evaluate AFPN on CUB-200-2011, Stanford Cars, and FGVC-Aircraft, achieving 89.7%, 94.2%, and 93.1% top-1 accuracy respectively, outperforming ResNet-50-FPN baselines by 2.1–3.4 percentage points with only 8% additional parameters. Ablation studies confirm that the attention gates contribute most at mid-level pyramid stages (P3–P4), where fine-grained texture details are most salient. Code and pretrained models will be released upon acceptance.

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1. Introduction

Fine-grained image classification aims to distinguish visually similar subcategories within a broader class, such as bird species, car models, or aircraft variants [1, 2]. The primary challenge lies in capturing subtle discriminative features—local texture patterns, part configurations, or color distributions—that differentiate closely related categories while remaining robust to intra-class variation in pose, scale, and background.

Feature Pyramid Networks (FPN) [3] have become a standard component in modern detection and classification architectures by constructing multi-scale feature representations through top-down pathways and lateral connections. However, the uniform fusion strategy in vanilla FPN treats all spatial locations and channels equally, ignoring the fact that discriminative fine-grained cues are often concentrated in specific regions and feature channels. Recent works have attempted to address this through part-based attention [4] or bilinear pooling [5], but these approaches typically operate on single-scale features and do not leverage the multi-scale hierarchy inherent in pyramid architectures.

In this paper, we introduce Adaptive Feature Pyramid Networks (AFPN), which embed a lightweight channel-spatial attention module at each lateral connection of the feature pyramid. Each attention gate receives the top-down feature map and the corresponding lateral feature map as dual inputs, producing a spatially and channel-wise modulated fusion output. This allows the network to dynamically upweight informative regions (e.g., bird head, car grille) at the pyramid level where their resolution is most appropriate for discrimination.

2. Method

Our architecture builds on a ResNet-50 backbone with standard FPN lateral connections from stages C2 through C5, producing pyramid levels P2–P5. At each lateral connection point, we insert an Attention Fusion Gate (AFG) module. The AFG takes two inputs: (1) the upsampled top-down feature map F_{td} and (2) the lateral feature map F_{lat} from the backbone. These are concatenated along the channel dimension and passed through a 1×1 convolution to produce joint attention weights.

The attention mechanism consists of a channel attention branch (global average pooling followed by two FC layers with ReLU and sigmoid) and a spatial attention branch (channel-wise max and average pooling followed by a 7×7 convolution with sigmoid). The two branches are multiplied element-wise to produce the final attention map $A \in R^{C \times H \times W}$. The fused output is computed as $F_{out} = A \otimes F_{lat} + (1 - A) \otimes F_{td}$, allowing the network to smoothly interpolate between bottom-up details and top-down semantics.

3. Experiments

We evaluate on three standard fine-grained benchmarks using standard train/test splits. Input images are resized to 448×448 with random horizontal flipping and color jittering during training. We use SGD with momentum 0.9, cosine annealing from 0.01 to 1e-5 over 90 epochs, and batch size 16. Table 1 summarizes the main results.

Table 1. *Top-1 accuracy (%) on fine-grained benchmarks.*

Method	CUB-200	Cars	Aircraft	Params
ResNet-50	84.5	90.3	89.2	25.6M
ResNet-50 + FPN	87.6	92.1	90.8	28.3M
ResNet-50 + SE-FPN	88.4	93.0	91.5	29.1M
AFPN (ours)	**89.7**	**94.2**	**93.1**	**30.5M**

4. Conclusion

We presented AFPN, a simple yet effective extension of Feature Pyramid Networks for fine-grained classification. By inserting lightweight attention gates at lateral connections, our method achieves consistent improvements across three benchmarks with minimal parameter overhead. Future work will explore extending AFPN to fine-grained retrieval tasks and investigating the attention patterns learned at each pyramid level to gain interpretability insights.

References

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